

BIOL3402 Cell Biology and Cell Technology

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Introduction

This course begins with a discussion of cell biology, covering the Cell, Membrane, Nuclear Transport, Techniques in Plant Cell Culture, Transport across the Plasma Membrane, Ion Channels and Osmotogenetics, Junction, Cytoskeleton, Cellular Adhesion molecules, Cell Migration, Cell Signaling and Techniques in Animal Cell Culture. This course aims to introduce the cellular knowledge. Since this is also a Lab-Lecture based course, this course also provide different culturing knowledge and laboratory skills. Upon completion, students will be familiar with the cell biology and cell technology conceptually and practically.

Contents

Introduction	1
1 Lecture 14 & 15 Cell Signaling	3

1 Lecture 14 & 15 Cell Signaling

1. Cell Signaling

- Conversion of extracellular signals into cellular responses
- A language of cells

2. Signaling Cascades (Transmission Chains)

- Composed of signal molecules
- Convert signals from external or internal stimuli
- Relay signal from point of reception to point of action in the cell
- Enlarge the received signal
- Distribute the signal to influence several responses in parallel
⇒ Long Term Reaction
- Each step may be regulated by other signals
⇒ E.g. Kinase Phosphate
- 1 Signaling → Multiple different expression form

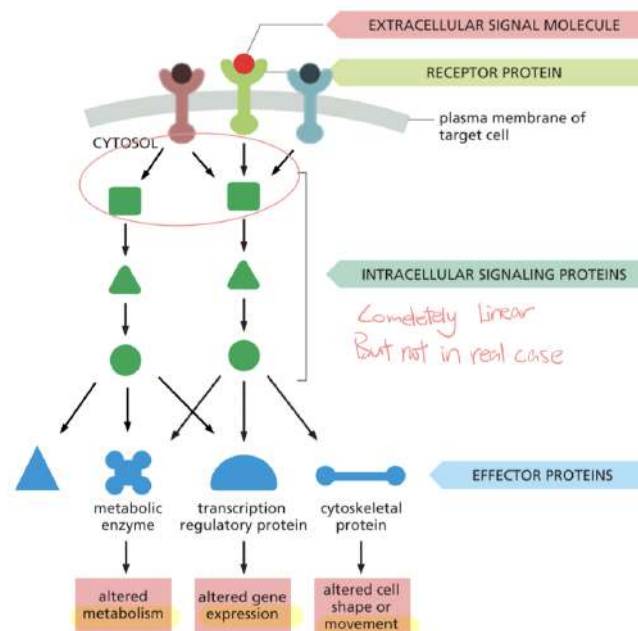


Figure 1: Signaling Cascades

- Multiple signals contribute to cell fate
 - (a) Survive
 - (b) Grow + Divide
 - (c) Differentiate
 - (d) Die → Apoptotic Cell

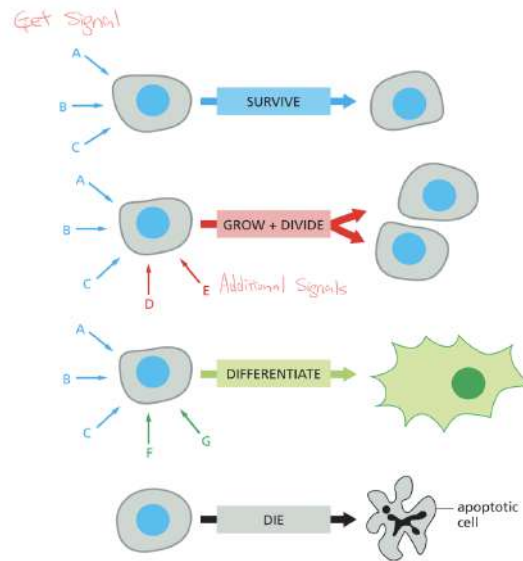


Figure 2: Cell Fate in different signal

- Produce diverse context-dependent
 - (a) Decrease rate of firing
 - (b) Secretion
 - (c) Contraction

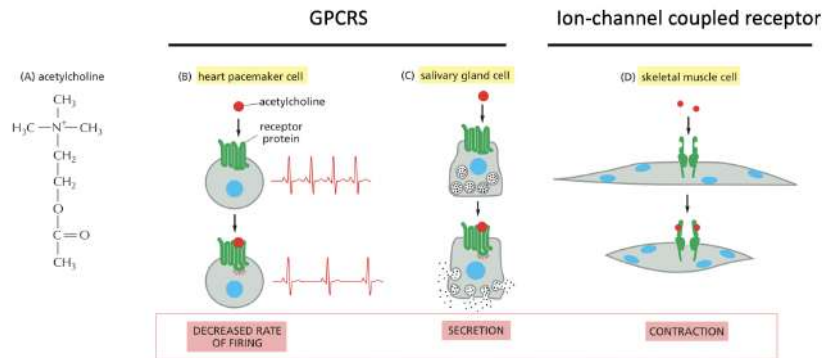


Figure 3: Context-dependent response

3. Types of Intercellular Signaling

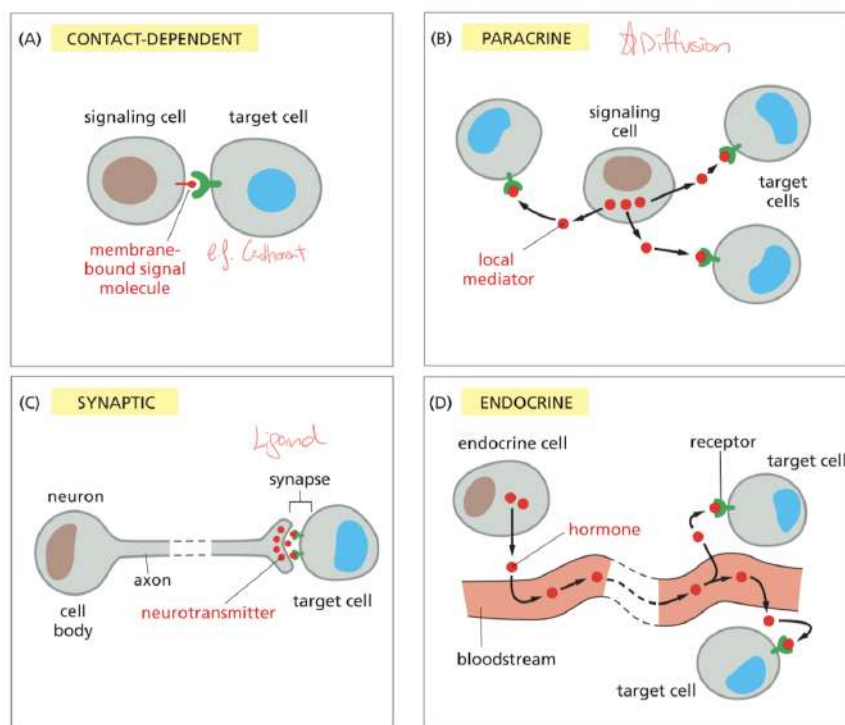


Figure 4: Intercellular Signaling

4. Speed of Extracellular Signals

- (a) Fast (Rapidly = Sec - Mins) : Altered Protein Function
 ⇒ E.g. Activate the G-Protein, Increase Heart Rate

(b) Slow (Mins - Hrs) : Altered Gene Expression (Usually happened in the nucleus)

⇒ E.g. Differentiation (Do not need to be fast), Make RNA, Folding, Making Protein, Transcription Fact Activation

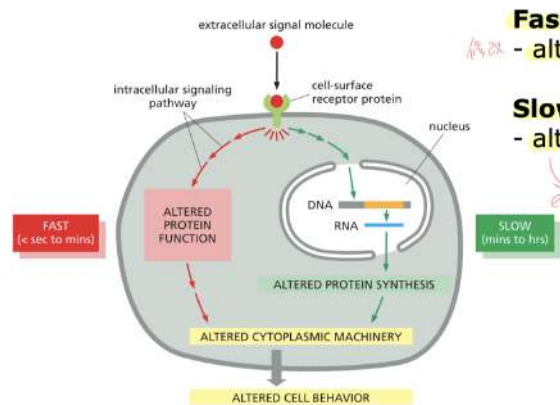


Figure 5: Speed of Extracellular Signal

5. Type of Extracellular Signaling Molecules

(a) Non Polar : Dissolve in the membrane

(b) Cell-Surface Receptors

- Receptor protein is on the surface of the cell
- Large or hydrophilic bind to membrane receptors
- A cargo of endocytosis
- Enter cell by the protein

(c) Intracellular Receptors

- Small or Hydrophobic pass through the membrane
- Can pass through membrane directly
- Receptor is in the nucleus

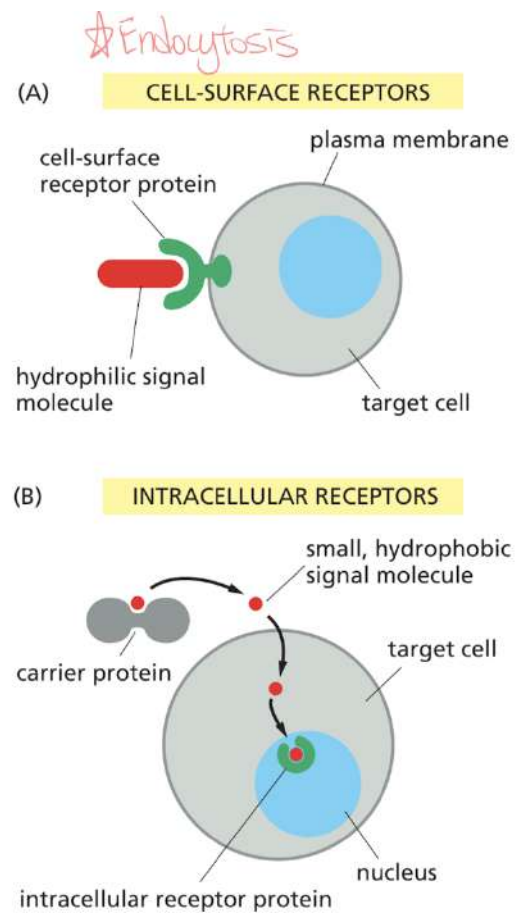


Figure 6: Extracellular Signaling Molecules

6. Type of Cell Surface Receptor

(a) Enzyme-coupled Receptors

⇒ E.g. Kinase

(b) G-protein-coupled Receptors

(c) Ion Channel Receptors

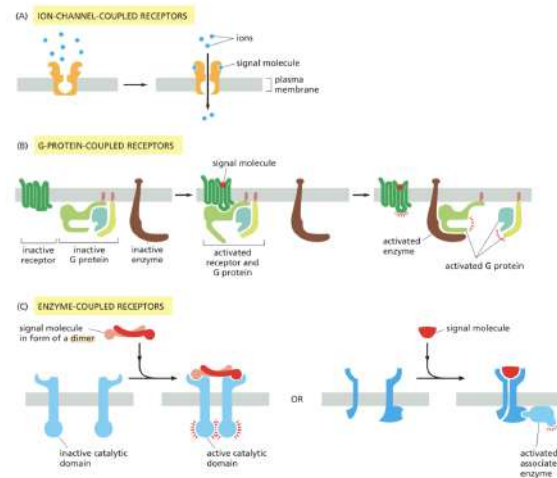


Figure 7: Cell Surface Receptor

7. Signaling Pathways

(a) Phosphorylation

(b) GTP Binding

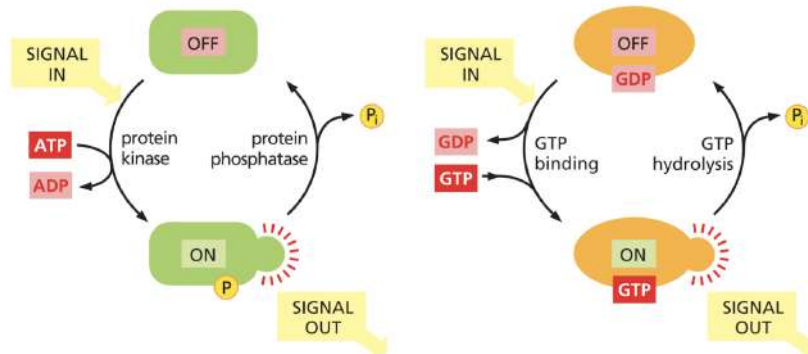


Figure 8: Signaling Pathways

8. GTPases : GTP is in 10-excess in the cytosol
 $\Rightarrow$ GTP is 10 times abundance than GDP

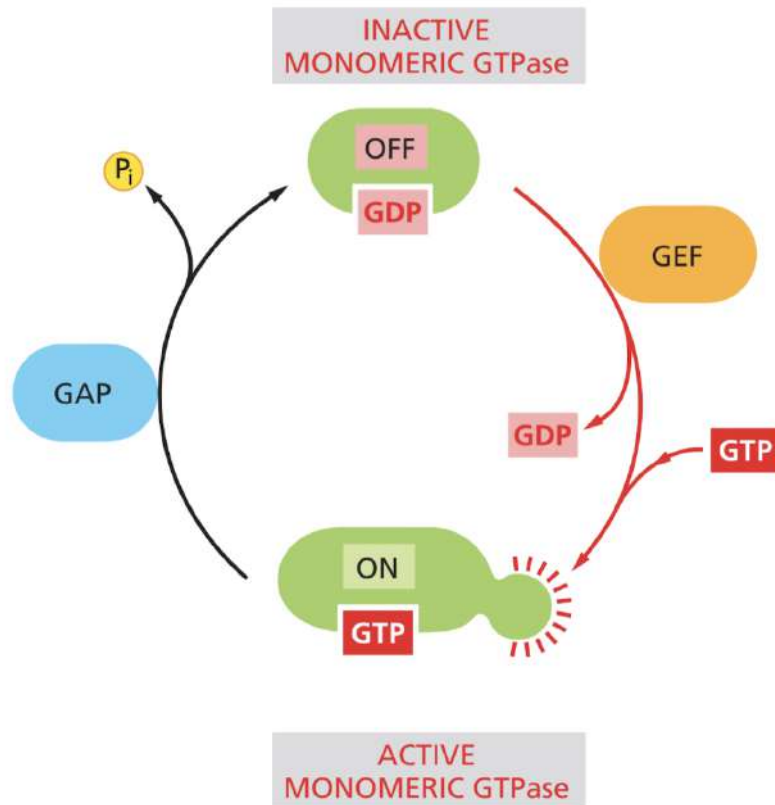


Figure 9: GTPases

9. Regulation of Signaling Pathways

- Positive Feedback
 - (a) Use to enhance the reaction
 - (b) If you want to stop it, it cannot stop
 - ⇒ Can only decrease a little bit of signal reactions
- Negative Feedback
 - (a) It will inhibit the reaction after one signal (Short Delay)
 - (b) Long delay will occurs and keep oscillating the signal reaction
 - (c) It can also be activating (Not only Inhibiting), but in negative regulator (Inhibitor Protein)
 - ⇒ Negative + Negative → Positive Feedback

- (d) Create Stable Patterns : Ensure a signal doesn't run out of control
- (e) Adapt : Return to baseline after being stimulated
- (f) Adjust Sensitivity : Become more or less sensitive to a signal over time

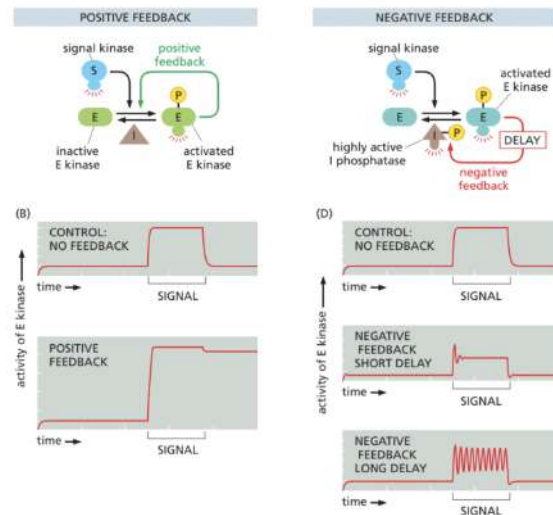


Figure 10: Regulation of Signaling Pathways

10. Feedbacks allow for switch-like response (Positive Feedback)
 ⇒ Once it open / start, it cannot stop (All-Or-None)

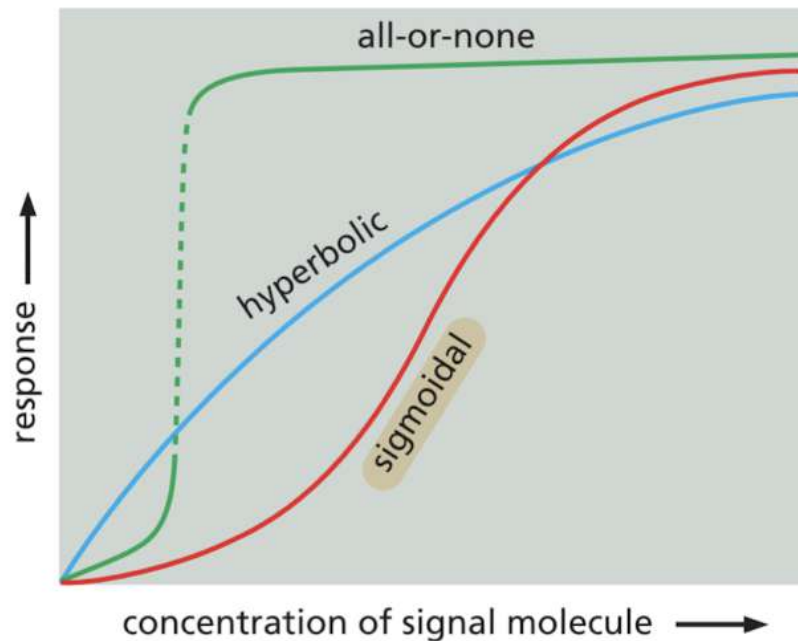


Figure 11: Feedbacks

11. Feedbacks allow for cells to adapt and adjust their sensitivity (Negative Feedback)

12. Negative Regulation + Negative Regulator

(a) Gene Expression in Closed State

- Inhibitor Protein's inactive transcription regulator is binded together
- Upstream Signal inactivate the Protein Kinase
- Block the transcription regulator + Activate the Inhibitor Protein with a Phosphate group
- Gene Expression is off

(b) Gene Expression in Open State

- Inhibitor Protein's inactive transcription regulator is binded together
- Upstream Signal activates the Protein Kinase
- Inhibit the Inhibitor Protein by Phosphorylation
- Activate the Transcription Regulator

- The transcription regulator is released and becomes active
- Gene Expression is opened
- Not just open and close system
- They are dynamic, interconnected systems
- Steps of Negative Feedback
 - (a) Activation : The researchers turn on the signal (MEK) in the middle
 - (b) Signal Propagation : This creates a strong local output of dpErk
 - (c) Feedback Loop is Triggered : The high level of dpErk (Or a downstream component) activates a mechanism that inhibits the signaling pathway itself

⇒ This could be by producing a phosphatase (An enzyme that removes phosphate groups) or by down regulating a receptor
 - (d) Global Dampening : This inhibitory signal spreads or acts globally, desensitizing the entire tissue

⇒ Even parts of the embryo that were not directly stimulated by light now have a higher threshold for activation
 - (e) Result : The endogenous signals at the poles are no longer strong enough to overcome this new, higher threshold

⇒ So their activity is depressed

13. Adaptation : Receptor can be recycled / dissolved

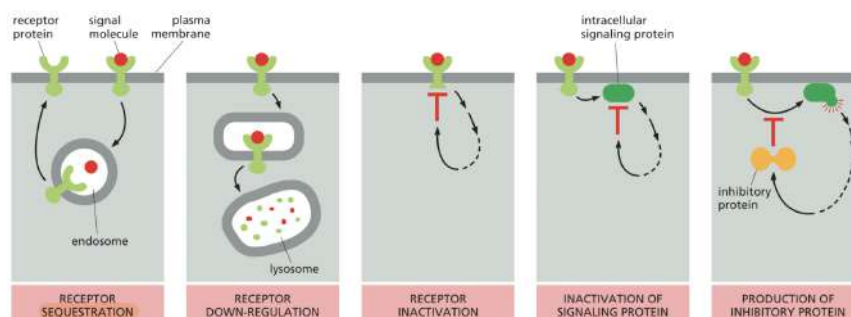


Figure 12: Mechanism of adaptation

14. Intracellular Signaling Complexes

- Localization is a type of recognition
- Type of complexes
 - (a) Assembly of signaling complex on an activated receptor
 - (b) Assembly of signaling complex on phosphoinositide docking sites
 - ⇒ The signal protein will only bind in correct position of specific phospholipid molecules (Specific Phosphoinositides)
 - ⇒ After Binding : Hyperphosphorylated Phosphoinositides
- Performed Signaling Complex on a **Scaffold Protein**
 - ⇒ Arrange the signaling Protein to prevent random signal diffusion

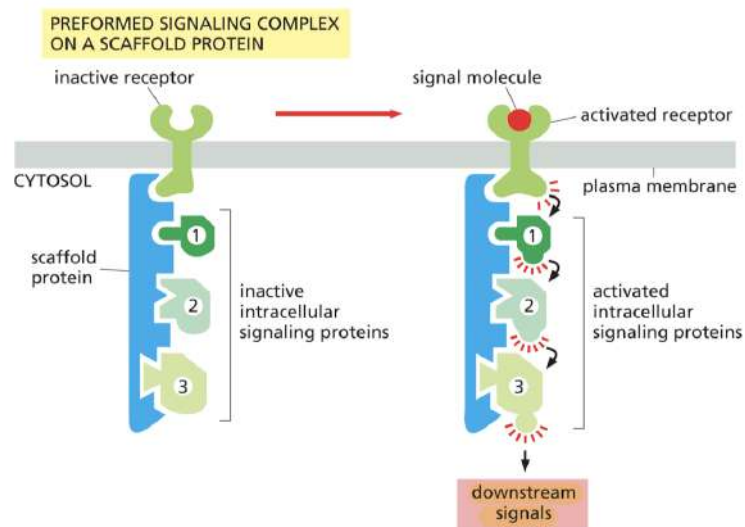


Figure 13: Intracellular Signaling Complexes (I)

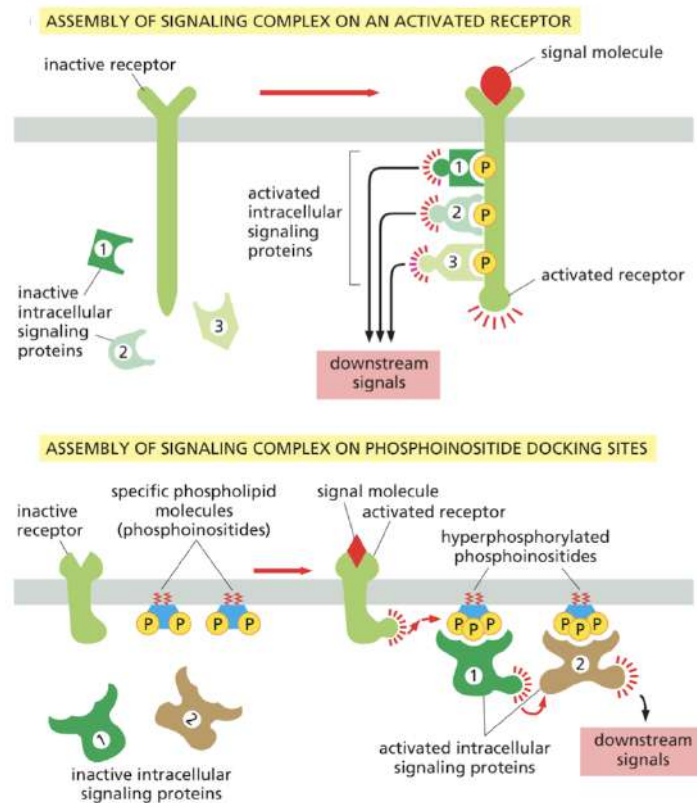


Figure 14: Intracellular Signaling Complexes (II)

15. Receptor Tyrosine Kinases (RTKs)

- All RTK ectodomains have highly variable structure : Bind to a ligand
- Transmembrane Domain on the plasma membrane : Anchorage
- All have a common cytoplasmic tyrosine kinase domain
- Cannot phosphorylate by itself
⇒ Only can phosphorylate with others
- Dysregulation of RTKs : Involve a tumor pathogenesis

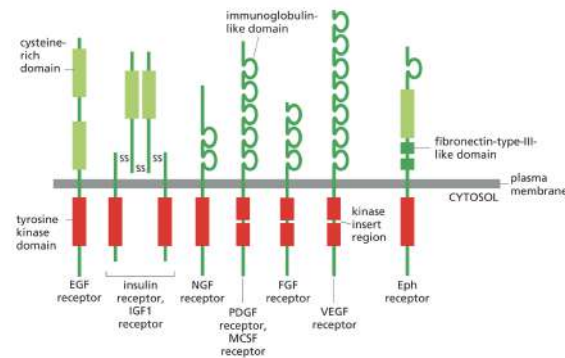


Figure 15: RTK Structure

- Steps of RTK Signaling
 - (a) Ligand bind to RTK
 - (b) RTK undergo conformational change
 - (c) Ligand-bound makes homodimers
 - (d) RTK homodimers phosphorylates each other
 - ⇒ This process called trans-autophosphorylation
 - ⇒ Activate the kinase domain
 - (e) The newly added phosphate groups on the tyrosine residues create binding sites for signaling proteins recognition
 - ⇒ Signaling proteins often have specific domains → Used for the binding to the phosphorylated tyrosines
 - (f) Signal Protein is activated
 - (g) Phosphorylate RTKs recruit downstream player
- Downstream Signaling : Happen inside the cell after a receptor has been activated by its initial signal

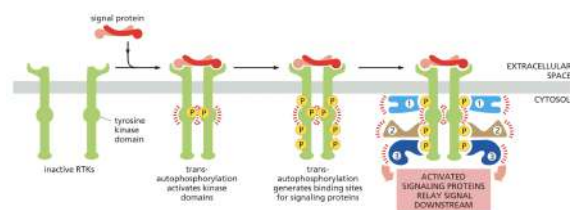


Figure 16: RTK Signaling

16. Ras signaling through RTK

- Ras : A small GTPase protein that acts as a crucial molecular switch
 ⇒ Activity : Controlled by whether it is bound to GDP (Off) or GTP (On)
- Steps of Ras Signaling
 - (a) A signal protein binds to the RTK, causing it to dimerize and auto-phosphorylate
 - (b) Adaptor protein (Grb2) recognizes the phosphorylated binding sites on the activated RTK → Binds to the RTK sites
 - (c) The Sos protein (Ras-GEF) get close to Ras
 - (d) Ras is initially in its inactive state, bound to GDP
 - (e) Sos (The Ras-GEF) forces Ras to throw GDP ∴ GTP is abundant in the cell, Ras immediately binds a new GTP molecule
 - (f) GTP causes a conformational change in Ras, shifting it to its active state
 - (g) Then it undergo the downstream signals
 - (h) Ras has intrinsic GTPase activity—it can hydrolyze GTP back to GDP
- If Ras gene mutated, then might cause cancer

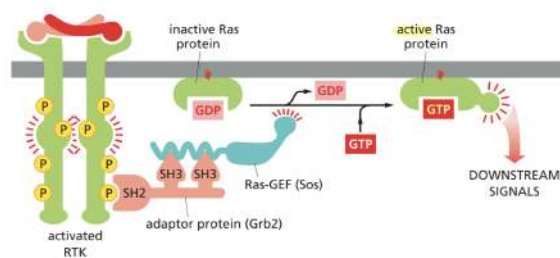


Figure 17: Ras Signaling

17. MAPK / ERK Signaling

- MAPK = Mitogen-Activated Protein Kinase
- ERK = Extracellular signal-Regulated Kinase : A specific type of MAPK

- Steps of MAPK / ERK Signaling
 - (a) Pathway begins with active Ras-GTP at the plasma membrane
 - (b) Ras activates MAP kinase kinase kinase (Raf)
 - ⇒ Raf is then activated by the binding
 - (c) Active Raf will undergo the double phosphorylation to 2 binding sites on MAP kinase kinase (Mek)
 - ⇒ It convert ATP to ADP during the binding
 - ⇒ MAP kinase kinase (Mek) is then activated
 - (d) Active Mek will undergo the double phosphorylation to 2 binding sites on MAP kinase (Erk)
 - ⇒ It convert ATP to ADP during the binding
 - ⇒ MAP kinase (Erk) is then activated
 - (e) Then covert a ATP to ADP again to do phosphorylation in cytoplasm / nucleus
 - Cytoplasm 1 : Phosphorylates other kinases and enzymes to immediately change cell metabolism and behavior
 - Cytoplasm 2 : Changes in protein activity
 - Cytoplasm 3 : Phosphorylates cytoskeletal proteins to alter cell shape and movement
 - Nucleus 1 : Change in gene expression
 - Nucleus 2 : Transcription factors turn on genes
 - ⇒ Promote cell cycle progression and cell growth
- Goal : Cell growth and division

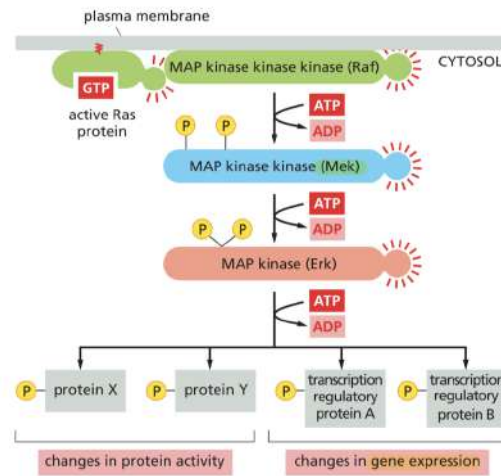


Figure 18: MAPK / ERK Signaling

18. Erk Signaling

- Same signaling pathway can trigger different cell fates
⇒ Depending on the dynamics of its activity
- EGF (Epidermal Growth Factor) : This stimulation leads to Cell Proliferation (The Cell Division)
⇒ Activation Time : 5 minutes
- NGF (Nerve Growth Factor) : This stimulation leads to Cell Differentiation
⇒ The cell stops dividing and matures into a specific type of neuron)
⇒ Activation Time : 45 minutes
- But both convert to Erk first
- Use different activation time to identify which pathways are going to do

19. PI3K Siganling

- Promotes Cell Survival + Growth + Migration
⇒ Method : Directly inhibiting programmed cell death (Apoptosis)
- In simple words, it transmits a **survival signal**
⇒ When active, it tells the cell that it is still survive

- Steps of PI3K Signaling
 - (a) A survival signal binds to an active receptor RTK
 - (b) Active RTK activates the central enzyme **PI 3-Kinase**
 - (c) PI3K phosphorylates a specific lipid in the plasma membrane called PIP_2
 - (d) Phosphorylation converts PIP_2 into PIP_3
 - (e) Two key kinases with PH domains binds with PIP_3 in **membrane**
 - $\Rightarrow$ PH Domain : PDK1 & Akt
 - (f) But Akt activation is depending to the PDK1 binding
 - (g) PDK1 phosphorylates Akt at first site + Another kinase mTOR phosphorylates Akt at a second site
 - (h) Active Akt dissociates from the membrane and moves into the cytosol
 - (i) Active Akt phosphorylates a protein "Bad"
 - $\Rightarrow$ Bad = A protein which promotes cell death
 - (j) Phosphorylated Bad is neutralized by a protein "14-3-3"
 - (k) The active apoptosis inhibitory protein that separate from Bad during phosphorylation : Inhibit the apoptosis $\rightarrow$ Cell Survive

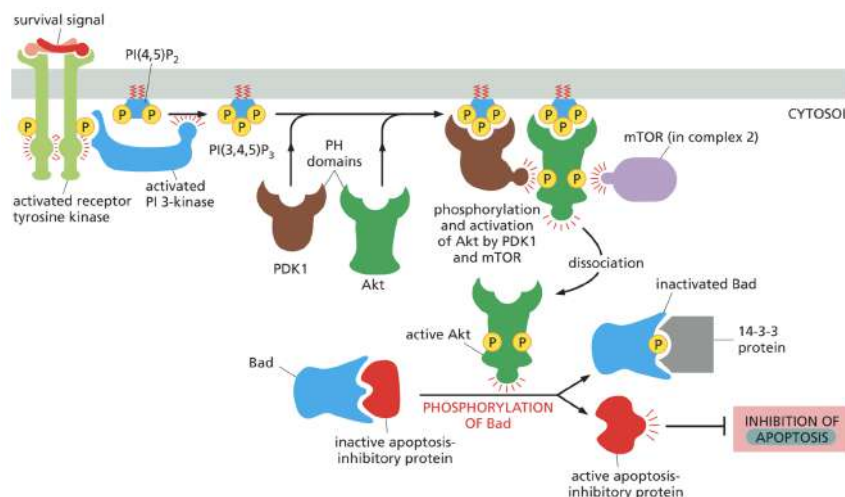


Figure 19: PI3K Signaling

- PI3K Signaling promotes cell migration

- Contact-dependent : Either one of the process and will-inhibit another one
- Rho Dominates : Actin -myosin contraction
- Rac Dominates : polymerization (Protrusion)
- Rho Dominates and Rac Dominates are contact dependent

20. G-protein Coupled Receptor

- Associated with G protein (Guanine-nucleotide binding protein) on cytoplasmic side
- Used in smell & vision : Regulate Ion Channels
- G protein are heterotrimers
 - ⇒ Heterotrimers = Protein complex composed of three separate polypeptide chains (subunits) that are different
- Contain 4 type of G Protein Subunit
 - ⇒ i.e. G_γ , G_β , G_α , $G_{\alpha\beta\gamma}$
- G_α & G_γ Subunit : Contain C-terminal cysteine residues and a prenyl group
 - ⇒ Can bind to form membrane lipid anchors
- G_α
 - Contain nucleotide binding and hydrolysis site + 2 Interactions Interfaces
 - ⇒ 2 Interaction Interfaces : GPCRs + $\beta\gamma$ Heterodimers
 - Active : GTP bounded
 - ⇒ Interact + Activate downstream signaling molecules
- $\beta\gamma$ Heterodimers
 - Stay tightly with each other
 - Can bind regulatory proteins + Downstream signaling molecules

21. α Subunits Function

- (a) G_{α_s} = Stimulates adenylyl cyclase + Regulate Ca^{2+} channels

- (b) $G_{\alpha i}$ = Inhibits adenylyl cyclase + Regulate K^+ & Ca^{2+} channels
- (c) $G_{\alpha q}$ = Activates phospholipase C_{β}
- (d) $G_{\alpha olf}$ = Olfactory (Smelling) + Stimulates adenylyl cyclase
- (e) $G_{\alpha t}$ = Transducin (Vision) + Activate cGMP phosphodiesterase

22. $\beta\gamma$ Subunit Function

- (a) $G_{\beta\gamma i}$ = Regulate Channels
- (b) $G_{\beta\gamma o}$ = Regulate Channels + Activate PLC_{β}

23. Steps of G-protein Coupled Receptor (GPCRs)

- (a) Signal Molecule Binding $\rightarrow$ Conformational Change $\rightarrow$ Creating the Activated GPCR
- (b) G Protein Activation : Inactive G protein is bound to the GPCR
- (c) GTP Binding and Subunit Dissociation
- (d) Downstream Signaling

24. G proteins regulate 2 type of targets

- Ion channel opening
- Membrane-bound enzymes
 - (a) Acetylcholine slows the heart
 - (b) Receptor activation
 - $\Rightarrow$ Dissociation of G_{α} and $G_{\beta\gamma}$
 - (c) $G_{\beta\gamma}$ opens K^+ channels
 - $\Rightarrow$ Decrease the amplitude of contraction

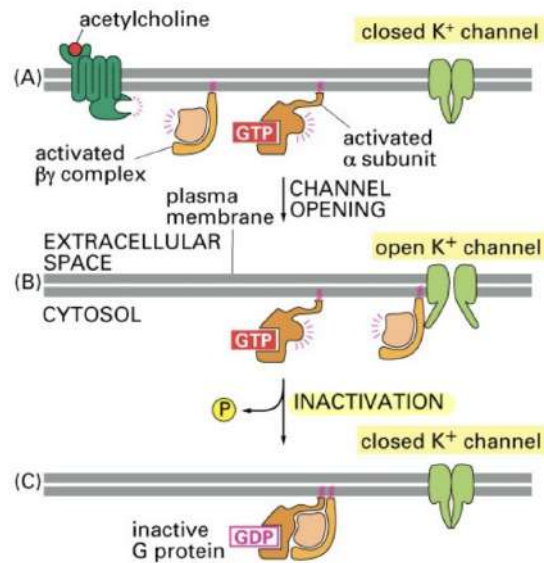


Figure 20: G Protein Regulation

25. Steps of Activated G Protein Dissociation

- Signal molecule binds GPCR
- Activated GPCR induces exchange of GDP for GTP on G_{α} subunit
- G_{α} dissociates from $G_{\beta\gamma}$
- Activated subunits diffuse within the plane of the membrane to activate downstream signaling molecules
- G_{α} can inactivate itself
 - $\Rightarrow$ By hydrolyzing its GTP
 - $\Rightarrow GTP \rightarrow GDP$
- Then, it forms heterotrimers again for next activation

26. Cyclic-AMP (cAMP) Activation through GPCRs

- Occurs in metabolic things
- Affect the transcription by the fast / slow respond
- Requires **Serotonin** + Little Time
- Activate PKA
 - $\Rightarrow$ Can enter nucleus and used for **gene transcription**

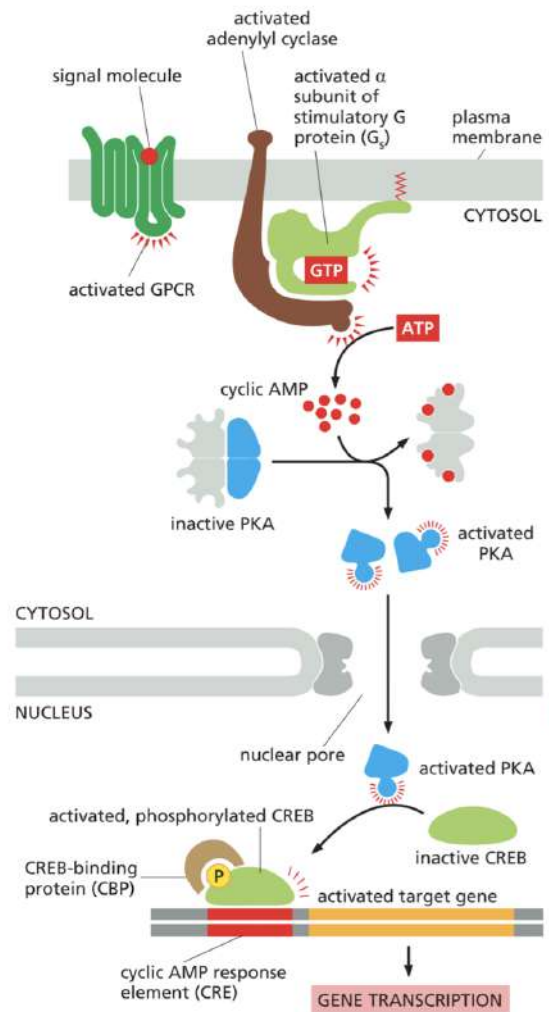


Figure 21: Cyclic-AMP (cAMP)

27. Steps of PLC / PKC (Protein Kinase C) Signaling

- (a) GPCR and G Protein Activation
- (b) Then activates a specific type of G protein called G_q
- (c) Activated G_q protein moves along the membrane + Activates the enzyme Phospholipase C-β
- (d) Activated PLC-β hydrolyzes a specific membrane phospholipid called PIP₂
⇒ Produce diacylglycerol + Inositol 1,4,5-trisphosphate (IP₃)
- (e) Split to two pathways

- IP_3 - Ca^{2+} Branch
 - i. IP_3 diffuses through the cytosol + Binds to IP_3 - gated Ca^{2+} channels on the surface of the Endoplasmic Reticulum (ER)
 - ii. The channel open, releasing a flood of Ca^{2+} ions from the ER lumen into the cytosol
 - The DAG Branch
 - i. DAG remains in the membrane and serves as a docking site to recruit and activate the enzyme Protein Kinase C (PKC)
- (f) Protein Kinase C (PKC) is normally inactive in the cytosol
 $\Rightarrow$ For it to become active, it requires two signals simultaneously
- i. Bind to DAG at the membrane
 - ii. Bind to the Ca^{2+} ions that have been released into the cytosol
 - iii. When both DAG and Ca^{2+} are present, PKC becomes fully activated
- (g) Activated PKC phosphorylates specific target proteins

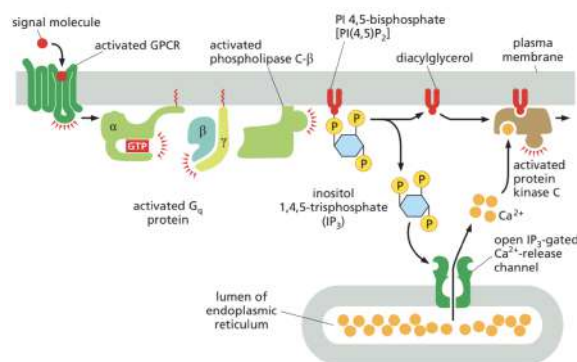


Figure 22: PLC / PKC Signaling

28. Calcium / Calmodulin Signaling : Control downstream kinase activity

- Many signals trigger Ca^{2+} release (Not only GPCRs)

- Skeletal muscle contracts in response to calcium release
- Ca^{2+} triggers regulated secretion (i.e. In Neurons)
- Sperm entry triggers a calcium wave during fertilization
- Ca^{2+} entry via plasma membrane Ca^{2+} channels and / or its release from the endoplasmic reticulum increases cytosolic Ca^{2+}

29. Calcium waves propagate through feedback

- IP₃ receptors : Need IP₃ + Ca^{2+}
- IP₃ binds to the ER gated ion channels
- Channel open and Ca^{2+} flow from ER lumen to cytosol
- IP₃ receptors and **Ryanodine Receptors** of the ER membrane : They are both stimulated by low to moderate cytoplasmic Ca^{2+} concentrations

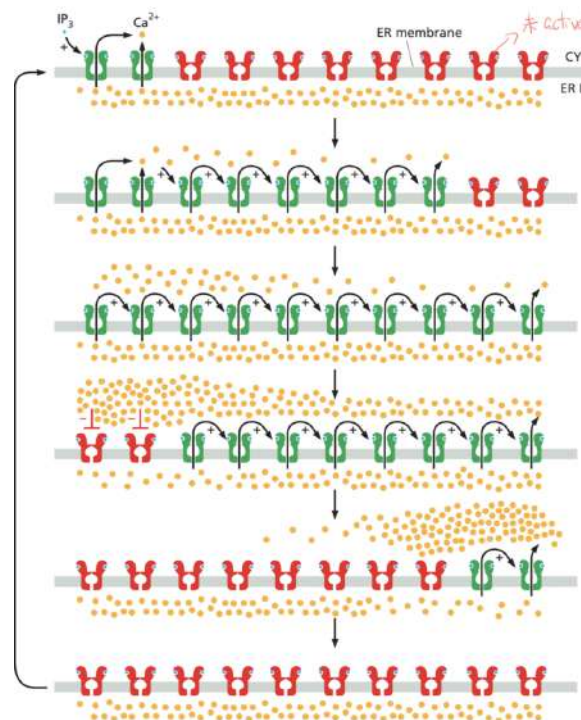


Figure 23: Calcium Waves

30. Signals are amplified by cascades

⇒ Rhodopsins : Light sensitive GPCRs

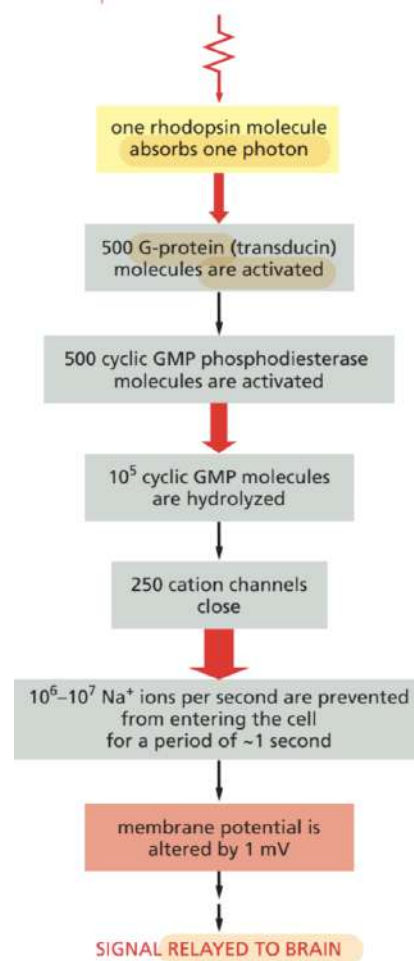


Figure 24: Signal to Brain

31. Signaling Pathways and Receptor is Cross-Talking

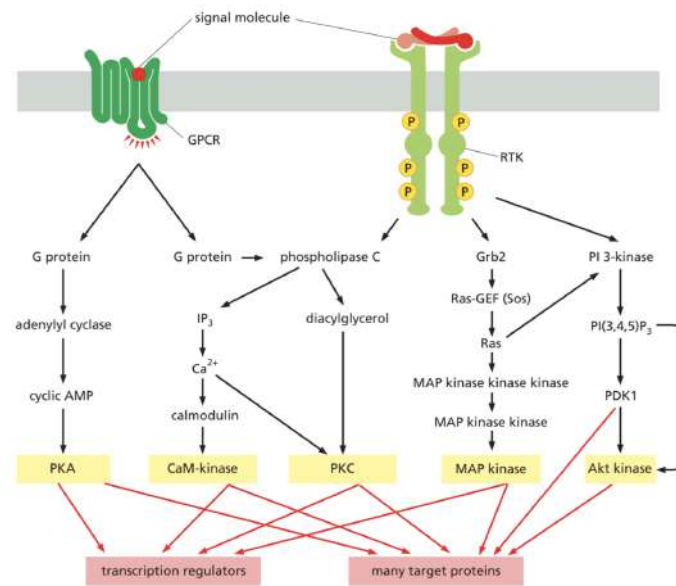


Figure 25: Cross-talking

32. JAK-STAT Signaling

- Only can phosphorylate with other Jaks
⇒ Similar to RTKs
- Activated by cytokines
- Cross Phosphorylation activates the JAK
- STAT Bind to SH2
- JAK generate a binding site and SH2 bind on it
- STAT is then enter nucleus
⇒ Gene Transcription

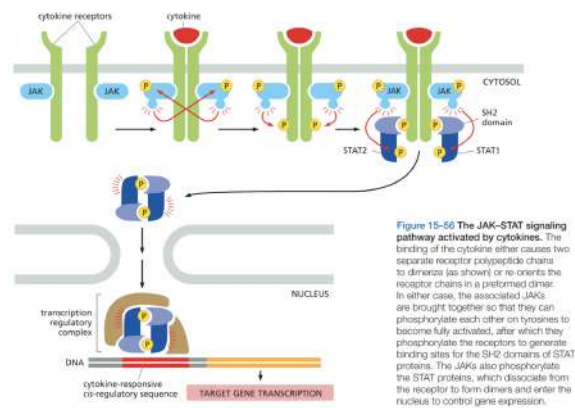


Figure 26: JAK-STAT Signaling